§3 Math for AI Safety — Worked Problems

Summary

3.1 Modular addition. A 1-layer transformer trained on 30% of $(a+b) \bmod 113$ memorises first (train loss < 0.1 by step 200) and generalises much later (test loss < 0.1 by step 8300, final test accuracy 99.9%). The learned embedding puts 95% of its norm on 5 Fourier frequencies and lays the residues out on circles. The structure forms before the test loss moves. Unlike in the paper, the **output** needs about 20 frequencies, not 5. **3.2 Open-source game theory.** An exact evaluator for provability-based agents (modal logic GL on Kripke chains) reproduces all 9 known outcomes, and gives DUPOC vs CUPOD = (D, C) for **unbounded** proofs. The open problem is the bounded case, which this does not settle.

1 Modular addition and the circular embedding (card 3.1)

Reproduces Nanda et al. (2023), *Progress measures for grokking via mechanistic interpretability*, in miniature. Code: `experiments/02_math/01_modular_addition.py` (CPU, about 40 min for 25000 steps).

1.1 Setup

Inputs are the three tokens "a b =" with $a, b \in \mathbb{Z}_p$ and $p = 113$. The model predicts $(a+b) \bmod p$ at "=". It is a 1-layer transformer (4 heads, $d_{\text{model}} = 128$, $d_{\text{mlp}} = 512$, ReLU, no LayerNorm, no biases), trained full-batch with AdamW (learning rate 10^{-3}, weight decay 1) on a random 30% of the p^2 pairs.

1.2 The algorithm the network is expected to find

With $\omega_k = 2\pi k/p$, the network can solve the task using only trigonometric identities. Embed each input on a circle, $a \mapsto (\cos \omega a, \sin \omega a)$, for a few **key frequencies** k. The MLP forms products, which by the angle-addition formulas give

$$\cos \omega(a+b) = \cos \omega a \cos \omega b - \sin \omega a \sin \omega b, \quad \sin \omega(a+b) = \sin \omega a \cos \omega b + \cos \omega a \sin \omega b. \tag{1}$$

The unembedding then scores each candidate answer c by

$$\text{logit}(c) \propto \sum_{k \in K} \cos \omega_k (a+b-c) = \sum_{k \in K} [\cos \omega_k (a+b) \cos \omega_k c + \sin \omega_k (a+b) \sin \omega_k c], \tag{2}$$

which is maximised exactly at $c \equiv a+b \pmod p$. Each frequency alone would also favour c values near $a+b$ on its own circle; summing several frequencies makes the true answer the unique peak.

To test this, expand the embedding $W_E \in \mathbb{R}^{p \times d}$ in the orthonormal real Fourier basis of $\mathbb{Z}_p$ and measure the share of $\|W_E\|_F^2$ carried by each frequency k (the cos and sin components together). The **restricted loss** projects the logits onto $\cos \omega_k (a+b-c)$ and $\sin \omega_k (a+b-c)$ for the top-5 frequencies only, and evaluates the test loss of that projection. It is low only if those few terms already implement the algorithm.

1.3 Results

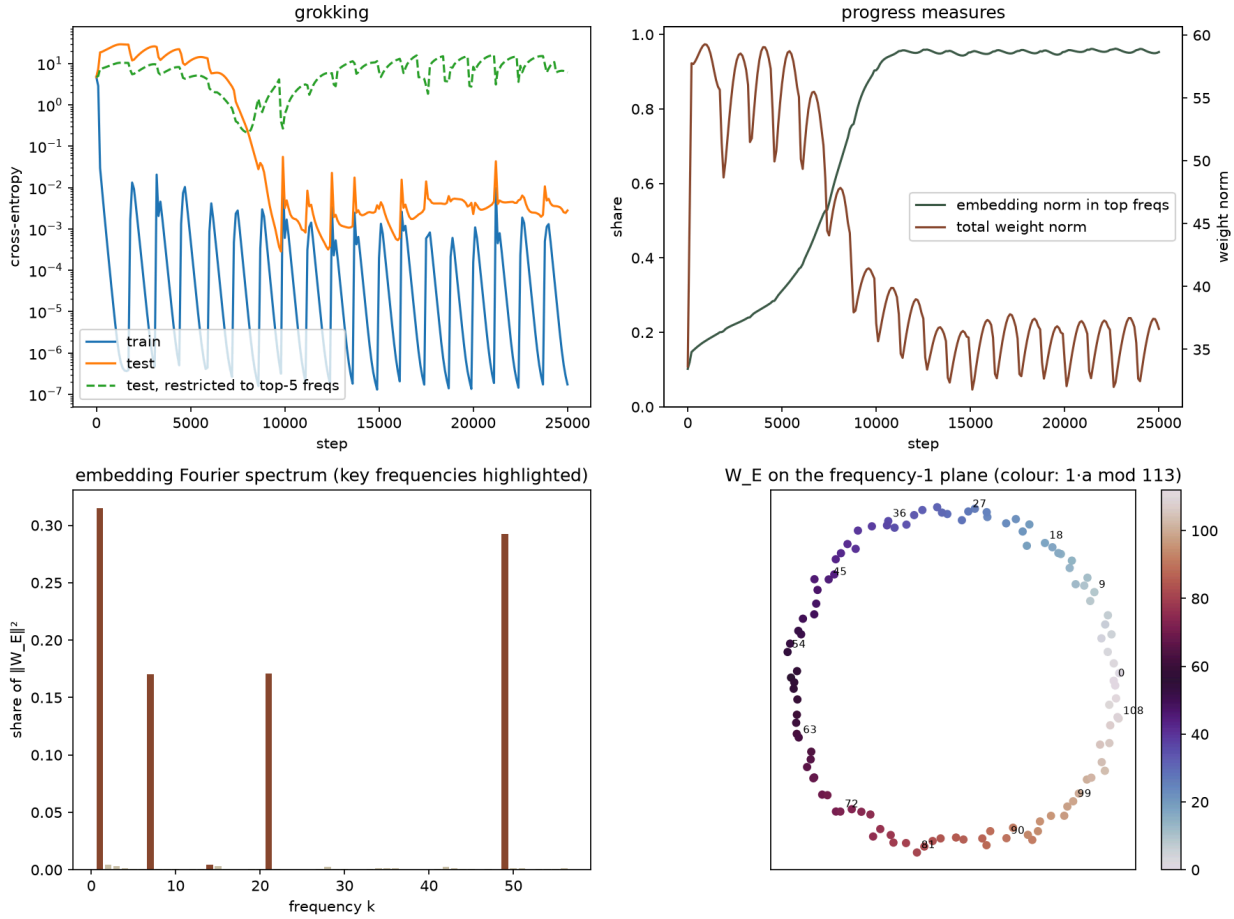


Figure 1: Top left: train, test and restricted test loss (log scale). Top right: share of the embedding norm in the top-5 frequencies, and the total weight norm. Bottom left: Fourier spectrum of the final embedding (key frequencies highlighted). Bottom right: the embedding projected on the plane of frequency 1, coloured by $1 \cdot a \bmod p$.

Finding

- **Grokking.** Train loss falls below 0.1 by step 200; test loss only by step 8300. Final test accuracy is 99.92%.
- **A sparse Fourier embedding.** The embedding puts 95% of its norm on frequencies 1, 49, 21, 7, 14. A random embedding would spread its norm evenly, about 9% on any 5 frequencies.
- **Circles.** On the plane of frequency 1, the 113 residues lie on a circle in their natural order a (for other key frequencies k the order is $ka \bmod p$). This is the "feature manifold" of the reading list's Interpretability & Feature Manifolds direction.
- **Neurons compute functions of $a + b$.** Median $R^2 = 0.93$, and 63% of the 512 neurons exceed 0.9.
- **The structure appears before generalisation.** The embedding's Fourier share rises steadily (Figure 1, top right) while test loss is still above 10 nats, and passes 0.9 at step 9900.

1.4 The output is not five clean cosines

Nanda et al.'s second progress measure, the **restricted loss**, keeps only the key frequencies of the logits. Here it does not behave as a progress measure: it dips at grokking and then rises to about 3 nats while the true test loss stays near 10^{-3} (Figure 1, dashed). I checked the implementation against an independent FFT version, and the two agree exactly, so this is a property of the model. Diagnostics on the final logits, taken as a function of the answer c (`experiments/02_math/01b_output_frequencies.py` for the sweep over K):

logits over c, test set	value
full model: test loss	0.0028
top frequencies of the logits (by power)	1, 49, 2, 3, 48
power share of those 5	90%
power share of the embedding's 5 key frequencies	77%
loss keeping only the logits' top 5	4.71
loss keeping only the embedding's 5 keys	6.08
loss **removing** the logits' top 5	29.2

Table 1: Where the logits' information lives. Uniform guessing is $\ln 113 \approx 4.73$ nats.

Finding

- **Power sits in low frequencies, which barely discriminate.** Frequency 1 alone carries 58% of the logit power, and $k = 1, 2, 3$ together carry 72%. A cosine of frequency 1 changes by only $1 - \cos(2\pi/113) \approx 0.0015$ of its amplitude between neighbouring answers.
- **About 20 frequencies are needed.** Keeping the top 5 by power (90% of it) leaves the model at chance: loss 4.7, argmax right 33% of the time. The top 10 give 89% accuracy, and the top 20 give 98% (loss 0.06).
- **Weak evidence for combination frequencies.** 10 of the top 20 logit frequencies are embedding keys or their pairwise sums and differences (1, 49, 2, 48, 15, 21, 50, 7, 6, 14), against 6.8 expected by chance ($p = 0.049$, 5,000 random draws). That is consistent with the MLP's ReLUs creating product terms that the unembedding reads, on top of the clean $\cos \omega(a + b - c)$ terms, but it's one model and a borderline test.

Caveat

- Nanda et al. trained for 40k steps; this run stopped at 25000. Weight decay keeps pruning after grokking, so the cleaner five-frequency output may simply need more training. That's cheap to test with `--steps 50000`.
- Two runs with the same seed found different key frequencies (1, 49, 21, 7, 14 here; 49, 3, 1, 21, 42 in an earlier run). Multi-threaded CPU arithmetic isn't bit-reproducible, and grokking amplifies tiny differences. Frequencies 1, 49 and 21 appeared in both runs. The circle, grokking and neuron findings held in both.

2 Open-source game theory with provability (card 3.2)

Code: `experiments/02_math/02_modal_agents.py`. Source: mathforaisafety.org, *Open-Source Game Theory*, and LaVictoire et al. (2014), *Robust Cooperation in the Prisoner's Dilemma*.

2.1 Agents as modal formulas

Two programs play a one-shot Prisoner's Dilemma, each able to read the other's source. An agent that decides by searching for **proofs** about the matchup is a formula of provability logic, where $\Box\varphi$ reads "PA proves φ". Writing $[X(Y)]$ for "X cooperates with Y":

$$[\mathrm{FairBot}(Y)] \leftrightarrow \Box[Y(\mathrm{FairBot})], \quad [\mathrm{Cupod}(Y)] \leftrightarrow \neg\Box\neg[Y(\mathrm{Cupod})], \tag{3}$$

$$[\mathrm{PrudentBot}(Y)] \leftrightarrow \Box[Y(\mathrm{PrudentBot})] \wedge \Box(\neg\Box\bot \rightarrow \neg[Y(\mathrm{DefectBot})]). \tag{4}$$

FairBot is DUPOC ("defect unless proof of cooperation"). CUPOD is "cooperate unless proof of defection".

Every agent variable appears under a $\Box$, so the system of equivalences has a unique solution (de Jongh–Sambin fixed-point theorem). Because GL is complete for PA's provability (Solovay), that solution can be computed exactly. Evaluate on the linear Kripke frame $0 < 1 < 2 < \ldots$, where $\Box\varphi$ holds at world n iff φ holds at every world $m < n$. World n then depends only on earlier worlds, the values become constant after finitely many worlds, and the constant value is the actual outcome. At world 0 every $\Box\varphi$ is vacuously true. That is the Löbian "benefit of the doubt" that lets FairBot cooperate with itself.

2.2 Results

row vs column	CooperateBot	DefectBot	FairBot	CUPOD	PrudentBot
CooperateBot	C	C	C	C	C
DefectBot	D	D	D	D	D
FairBot	C	D	C	D	C
CUPOD	C	D	C	D	C
PrudentBot	D	D	C	D	C

Table 2: The row agent's move against the column agent (unbounded proofs). Computed, not assumed.

Finding

- **All 9 known outcomes reproduce**, including the subtle ones. PrudentBot exploits CooperateBot, while FairBot does not. CUPOD defects against **itself**, even though it seems the "nicer" bot: $p \leftrightarrow \neg\Box\neg p$ has the unique solution $p \leftrightarrow \bot$.
- **DUPOC vs CUPOD = (D, C)** in the unbounded setting, stable from world 1. At world 0, DUPOC cooperates (vacuous $\Box$), and so $\neg\Box\neg$ fails for CUPOD, which defects. From world 1 on, $\Box$[CUPOD cooperates] is false because of world 0, so DUPOC defects. $\neg\Box\neg$[DUPOC cooperates] is then true because DUPOC cooperated at world 0, so CUPOD cooperates. This is the GL version of the page's argument: PA can't prove that CUPOD cooperates with anyone, so DUPOC defects, and CUPOD, finding no proof of defection, cooperates.

Caveat

- **This does not settle the open problem.** The problem is about **bounded** proofs, $\Box_k$ ("a proof of length $\leq k$"), where GL's axioms fail. The known bounded results (DUPOC–DUPOC cooperate, CUPOD–CUPOD defect) need the parametric bounded Löb theorem, and the obstacle cited for DUPOC(k) vs CUPOD(k) is the fixed point $X \leftrightarrow \neg\Box_k\neg\Box_k X$. The GL answer is evidence for "(D,C) eventually", not a proof.
- **A likely erratum on the source page.** It says "both CUPOD and DUPOC cooperate against FairBot" but also defines FairBot as DUPOC, and derives that DUPOC **defects** against CUPOD. Those claims can't all hold. In the table, CUPOD cooperates with FairBot but FairBot defects against CUPOD.

Next (your original work): a toy **bounded** proof search, for example a small proof system with explicit proof-length bounds where $\Box_k$ is decided by search, checked first against the known small-k outcomes above, and then run on DUPOC(k) vs CUPOD(k) as k grows.

3 No-coincidence argument (card 3.3)

Original card. The statement of the conjecture, the exercise and a starting point are in `experiments/02_math/03_no_coincidence.md`. The network from 3.1 is a natural first candidate: it is correct on all p^2 inputs, which for random weights would be a coincidence of probability about p^{-p^2}, and the Fourier circuit above is its structural explanation.

4 Status of §3

DONE 3.1 Modular addition: grokking, Fourier embedding, circles, neurons, and an output-side analysis of the restricted-loss discrepancy.

IN PROGRESS 3.2 Modal agents: exact unbounded evaluator and known results done. Bounded proof search is yours.

TO DO 3.3 No-coincidence mini-argument: scaffold in `03_no_coincidence.md`.